

PIZZA SALES DATA ANALYSIS USING SQL

A Delicious Journey



All
About
Pizza!

BASIC

1. Retrieve the total number of orders placed.

```
SELECT COUNT(order_id) AS total_orders
FROM orders;
```

2. Calculate the total revenue generated from pizza sales.

```
SELECT ROUND(SUM(order_details.quantity *
pizzas.price), 2) AS total_sales
FROM order_details
JOIN pizzas
ON pizzas.pizza_id = order_details.pizza_id;
```

3. Identify the highest-priced pizza.

```
SELECT pizza_types.name, pizzas.price
FROM pizza_types
JOIN pizzas
ON pizza_types.pizza_type_id = pizzas.pizza_type_id
ORDER BY pizzas.price DESC
LIMIT 1;
```

4. Identify the most common pizza size ordered.

```
SELECT pizzas.size,
COUNT(order_details.order_details_id) AS order_count
FROM pizzas
JOIN order_details
ON pizzas.pizza_id = order_details.pizza_id
GROUP BY pizzas.size
ORDER BY order_count DESC
LIMIT 1;
```

5. List the top 5 most ordered pizza types along with their quantities.

```
SELECT pizza_types.name,
SUM(order_details.quantity) AS quantity
FROM pizza_types
JOIN pizzas
ON pizza_types.pizza_type_id = pizzas.pizza_type_id
JOIN order_details
ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY quantity DESC
LIMIT 5;
```


INTERMEDIATE

6. Join the necessary tables to find the total quantity of each pizza category ordered.

```
SELECT pizza_types.category,  
SUM(order_details.quantity) AS quantity  
FROM pizza_types  
JOIN pizzas  
ON pizza_types.pizza_type_id = pizzas.pizza_type_id  
JOIN order_details  
ON order_details.pizza_id = pizzas.pizza_id  
GROUP BY pizza_types.category  
ORDER BY quantity DESC;
```

7. Determine the distribution of orders by hour of the day.

```
SELECT HOUR(order_time) AS hour,  
COUNT(order_id) AS order_count  
FROM orders  
GROUP BY HOUR(order_time)  
ORDER BY hour;
```

8. Join relevant tables to find the category-wise distribution of pizzas.

```
SELECT category,  
COUNT(name) AS pizza_count  
FROM pizza_types  
GROUP BY category;
```

9. Group the orders by date and calculate the average number of pizzas ordered per day.

```
SELECT ROUND(AVG(quantity), 0) AS avg_pizza_ordered_per_day  
FROM  
(  
    SELECT orders.order_date,  
    SUM(order_details.quantity) AS quantity  
    FROM orders  
    JOIN order_details  
    ON orders.order_id = order_details.order_id  
    GROUP BY orders.order_date  
)  
AS order_quantity;
```

10. Determine the top 3 most ordered pizza types based on revenue.

```
SELECT pizza_types.name,  
SUM(order_details.quantity * pizzas.price) AS revenue  
FROM pizza_types  
JOIN pizzas  
ON pizza_types.pizza_type_id = pizzas.pizza_type_id  
JOIN order_details  
ON order_details.pizza_id = pizzas.pizza_id  
GROUP BY pizza_types.name  
ORDER BY revenue DESC  
LIMIT 3;
```

ADVANCED

11. Calculate the percentage contribution of each pizza type to total revenue.

```
SELECT pizza_types.category,
ROUND(
SUM(order_details.quantity * pizzas.price) /
(SELECT ROUND(SUM(order_details.quantity * pizzas.price), 2)
FROM order_details
JOIN pizzas
ON pizzas.pizza_id = order_details.pizza_id) * 100,
2
) AS revenue
FROM pizza_types
JOIN pizzas
ON pizza_types.pizza_type_id = pizzas.pizza_type_id
JOIN order_details
ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY revenue DESC;
```

12. Analyze the cumulative revenue generated over time.

```
SELECT order_date,
SUM(revenue) OVER (ORDER BY order_date) AS cumulative_revenue
FROM
(
SELECT orders.order_date,
SUM(order_details.quantity * pizzas.price) AS revenue
FROM order_details
JOIN pizzas
ON order_details.pizza_id = pizzas.pizza_id
JOIN orders
ON orders.order_id = order_details.order_id
GROUP BY orders.order_date
) AS sales;
```

13. Determine the top 3 most ordered pizza types based on revenue for each pizza category.

```
SELECT category,
name,
revenue
FROM
(
SELECT category,
name,
revenue,
RANK() OVER(PARTITION BY category ORDER BY revenue DESC) AS rn
FROM
(
SELECT pizza_types.category AS category,
pizza_types.name AS name,
SUM(order_details.quantity * pizzas.price) AS revenue
FROM pizza_types
JOIN pizzas
ON pizza_types.pizza_type_id = pizzas.pizza_type_id
JOIN order_details
ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category, pizza_types.name
) AS a
) AS b
WHERE rn <= 3;
```

THANK
YOU

